

Group members:

1. Lau Yee Wen A23CS0099
2. Gui Kah Sin A23CS0080

SECJ2013 - DSA

2024/2025-1

QUIZ 1

Given that two functions (**funA** and **funB**) have been defined as follows:

```
void funA(int n) {  
    cout << "funA - " << n << "\n";  
    n = n + 1;  
    if (n < 5) {  
        if (n % 2 == 0) {  
            funB(n);  
        } else {  
            funA(n);  
        }  
    }  
}
```

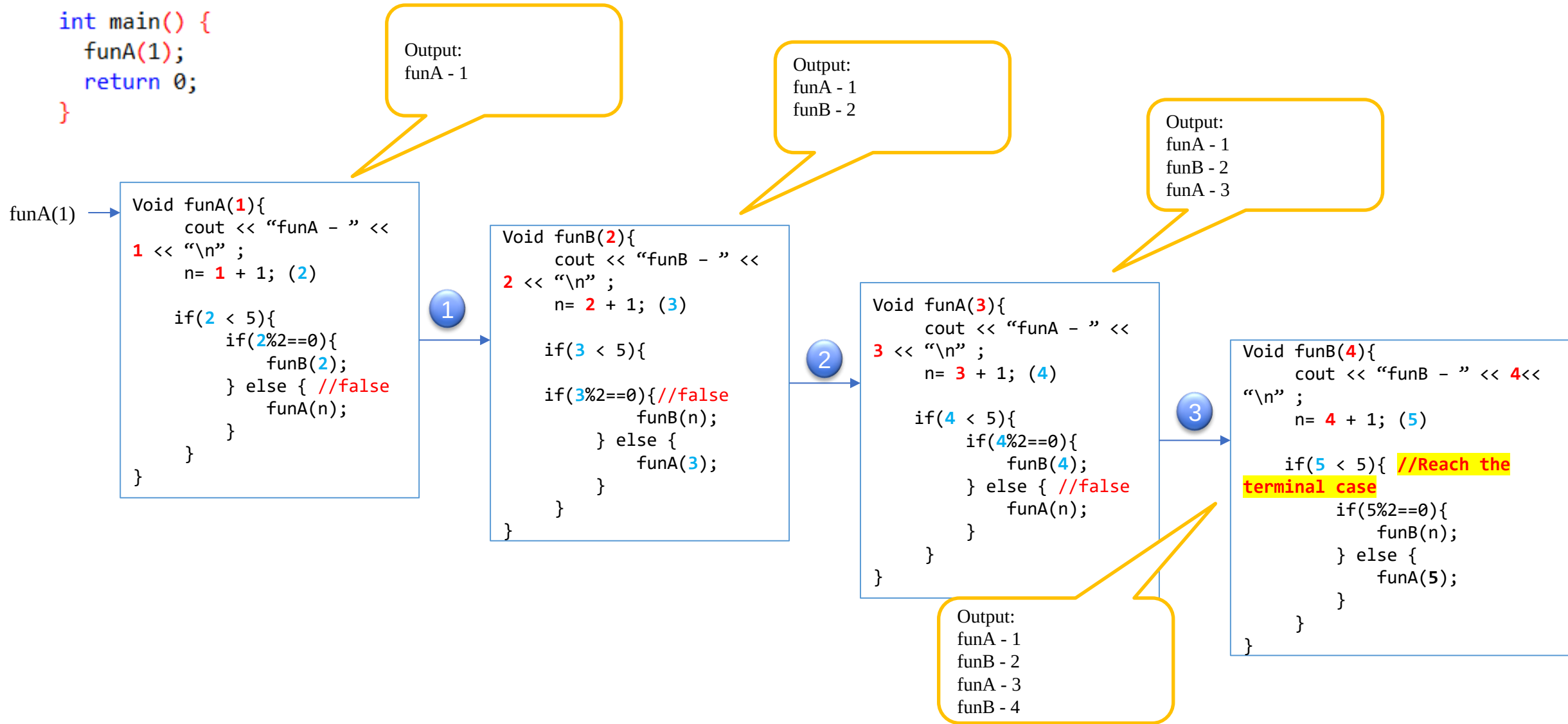
```
void funB(int n) {  
    cout << "funB - " << n << "\n";  
    n = n + 1;  
    if (n < 5) {  
        if (n % 2 == 0) {  
            funB(n);  
        } else {  
            funA(n);  
        }  
    }  
}
```

Study the funA and funB functions above. Identify the followings:

1. Instruction to check if the terminal case has been reached: if(n < 5)
2. Instruction to change the size of the problem so the terminal case can be reached: n = n + 1

Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```
int main() {  
    funA(1);  
    return 0;  
}
```



Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```
int main() {  
    funA(1);  
    return 0;  
}
```

